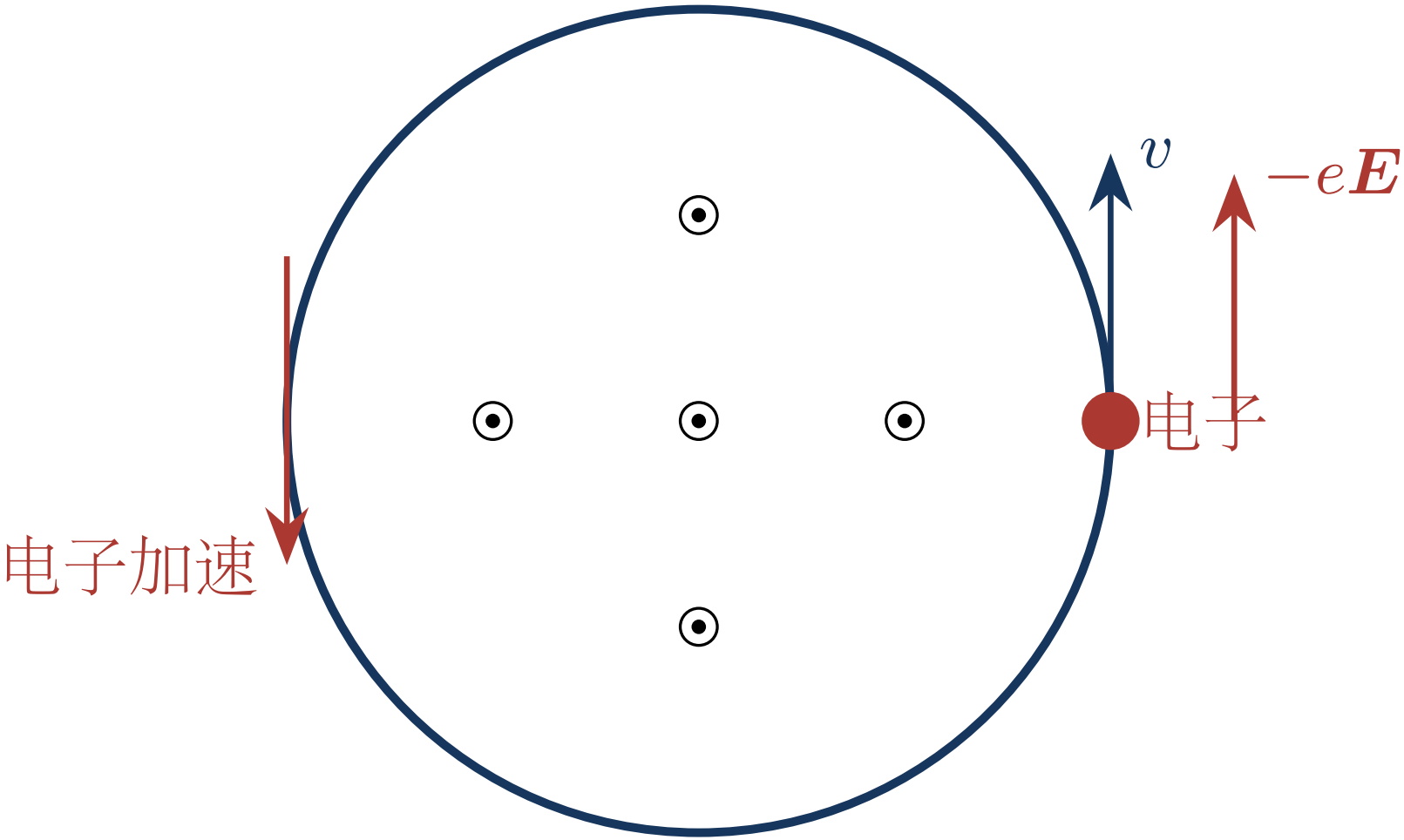
$$B > 0,\ \dot{B} > 0$$



$$E_\varphi = -r\dot{B}/2$$